Reproduction of Quantum Key Distribution Based on E91 Protocol

Course Project Report

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Agenda

- Introduction: Project Objectives

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- Background: Quantum Cryptography & Key Protocols

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- Methodology: E91 Reproduction & BB84 Comparison

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- Q & A

Classical Key Distribution: A Critical Dilemma

Classical Communication Channel

- Keys must be shared via a "secure channel"
- Passive eavesdropping is *undetectable*
- Security relies on computational complexity (e.g., RSA: integer factorization)

Quantum Communication Channel

- Keys are generated *during transmission*
- Eavesdropping disturbs quantum states (physical trace)
- Security relies on *quantum mechanics laws* (unconditional security)

Two Foundational QKD Protocols

Protocol	Proposed (Year)	Core Physical Principle
BB84	Bennett & Brassard (1984)	Heisenberg Uncertainty Principle -
E91	Ekert (1991)	Quantum Nonlocality (Bell's Theo

Table: Key Differences of Two QKD Protocols

Project Motivation:
Reproduce E91 protocol, verify its unconditional security, and compare with BB84 to clarify their application scenarios.

Project Design & Tools

Core Objectives:

1. Reproduce E91 protocol (key generation + eavesdropping detection)

Tools & Environment:

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- Experimental Reference: Aspect's Bell test experiment (1982)

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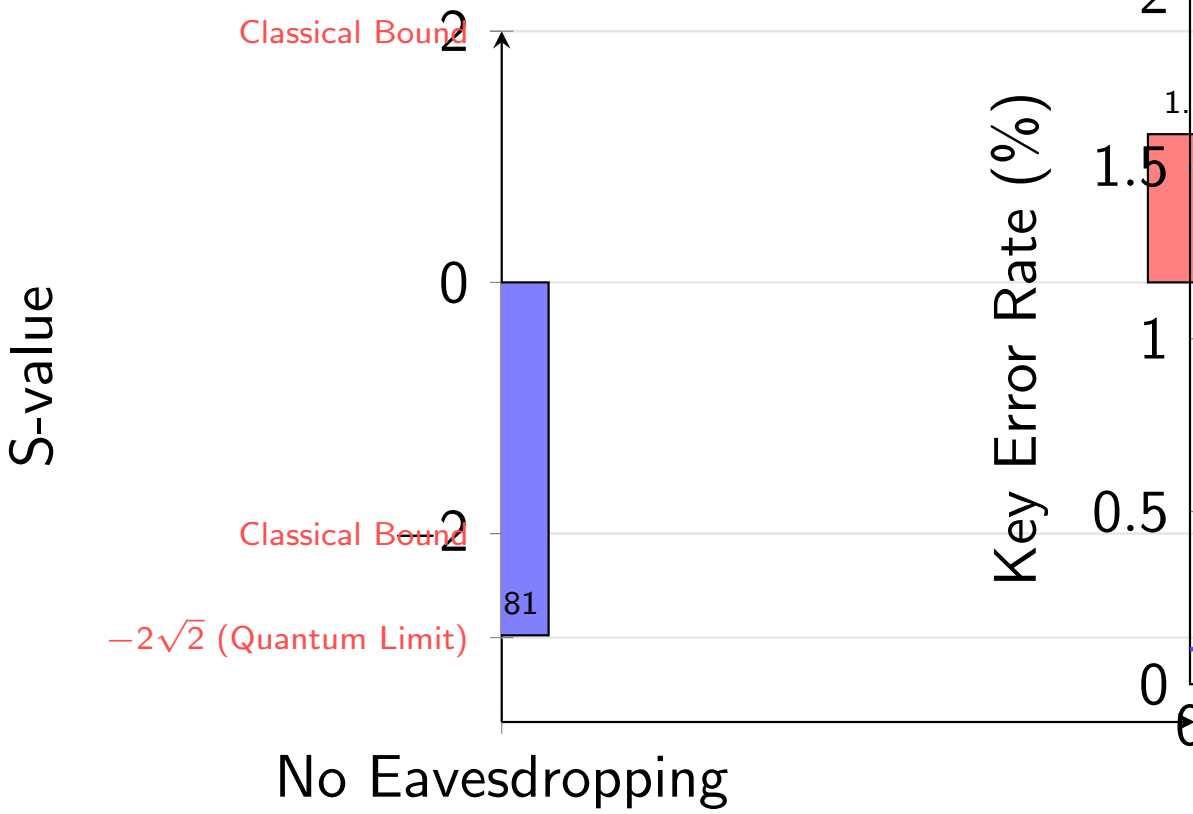
5. , label=Step 5:**Key Generation:** Same-orientation results are fully anticorrelated $\rightarrow$ Alice keeps raw values, Bob inverts to get identical keys

E91 vs. BB84: Comparative Framework

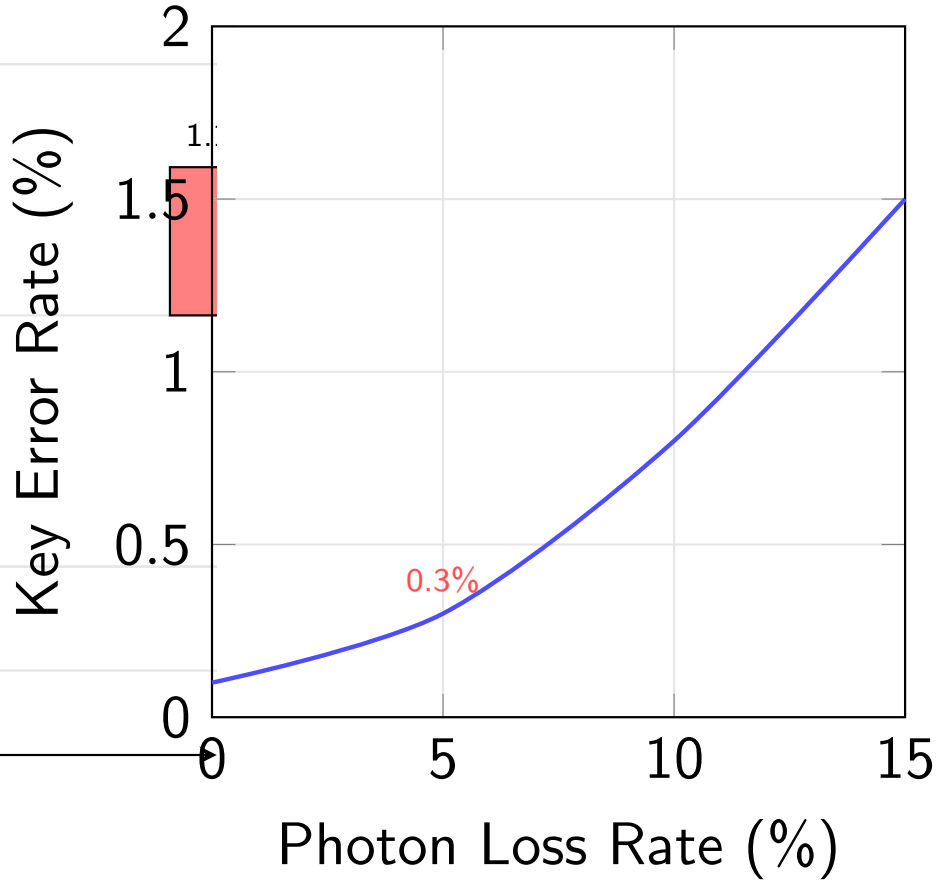
Dimension	**E91 Protocol**
Physical Foundation	Quantum Nonlocality (Be
Experimental Complexity	High (requires entangled
Key Efficiency	Higher (no basis-discardin
Robust against photon-number splitting attacks	Dependent on single-phot

Comparison Method: Test both protocols under the same conditions (5

E91 Protocol: Key Results



S-value Comparison



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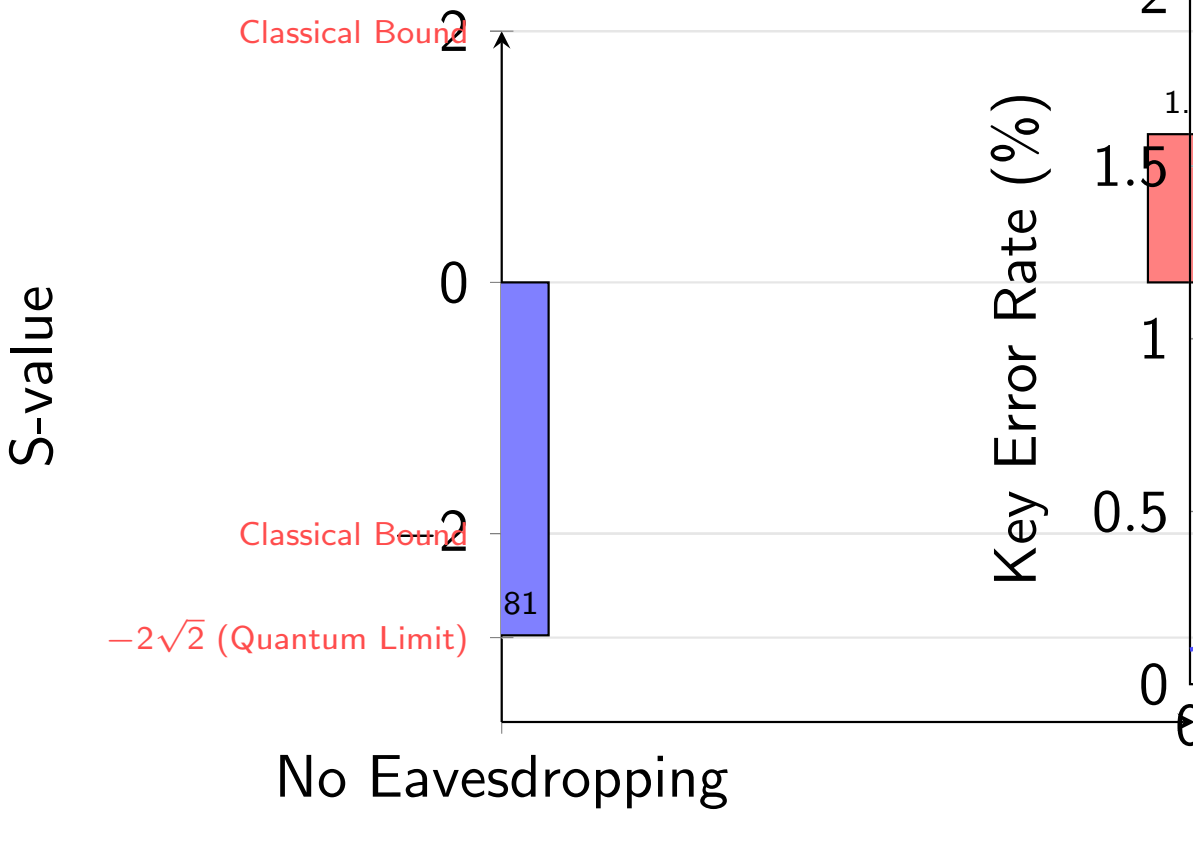

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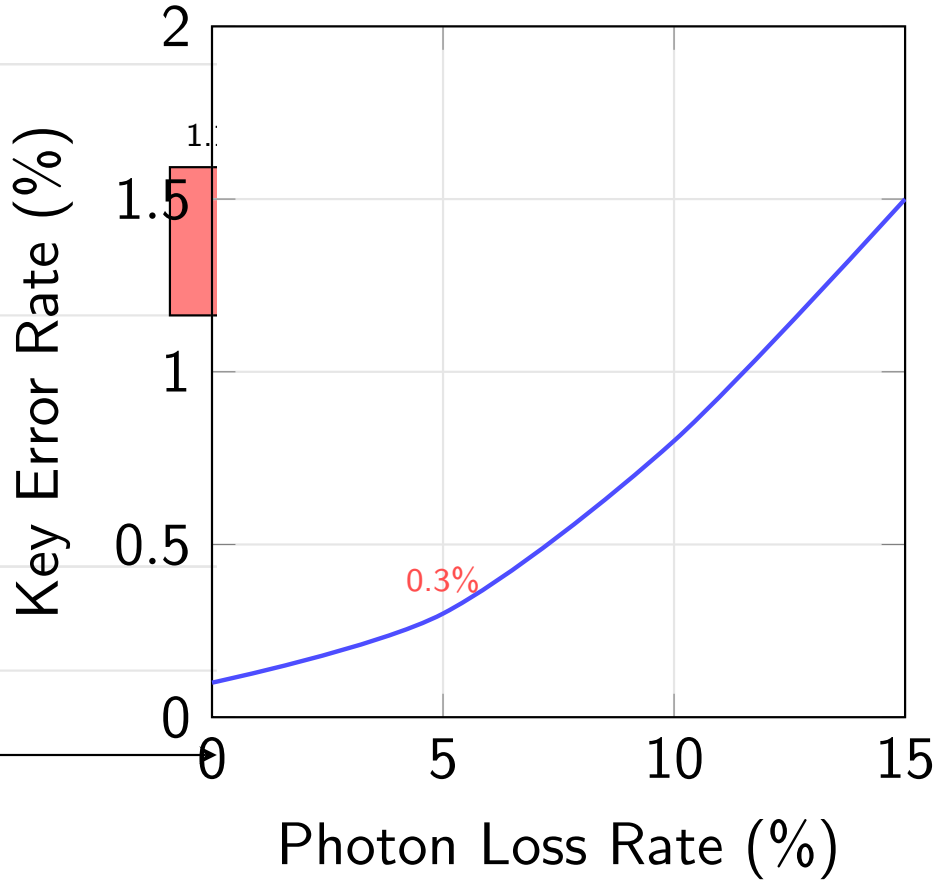

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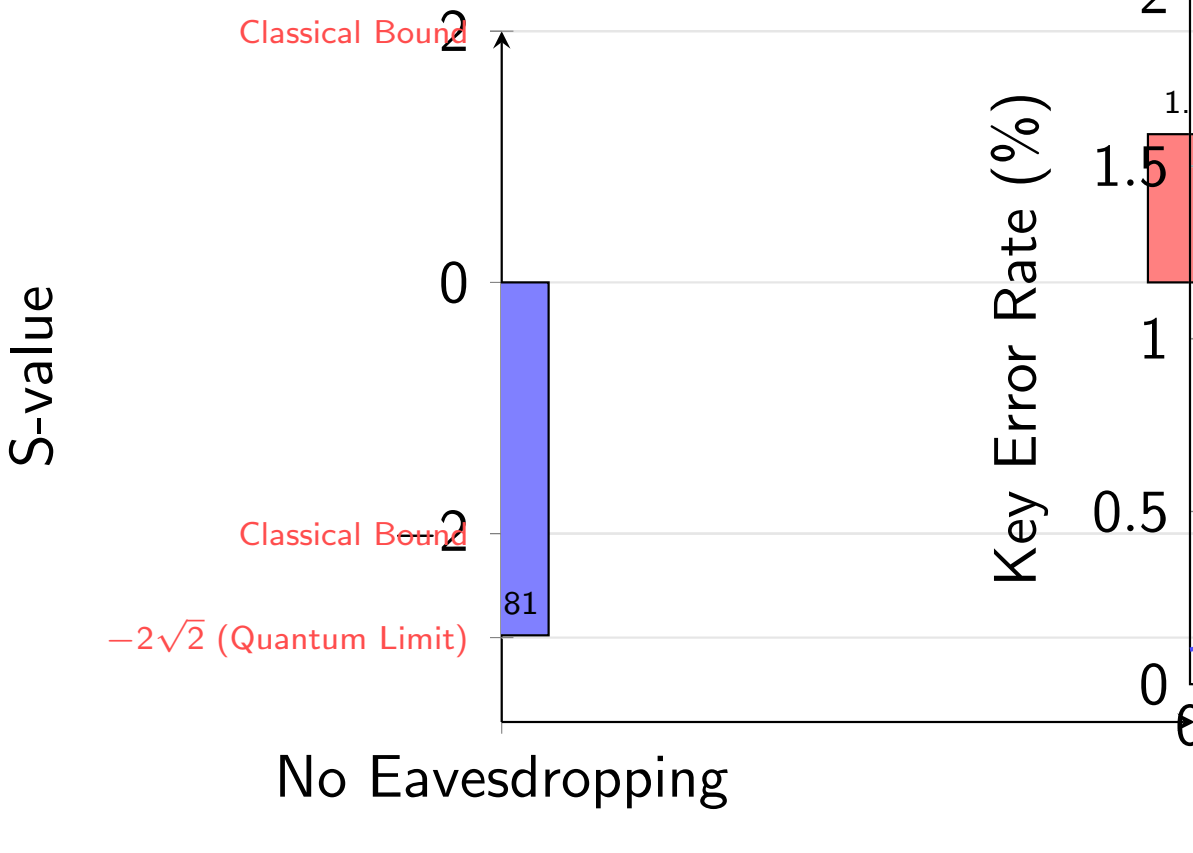

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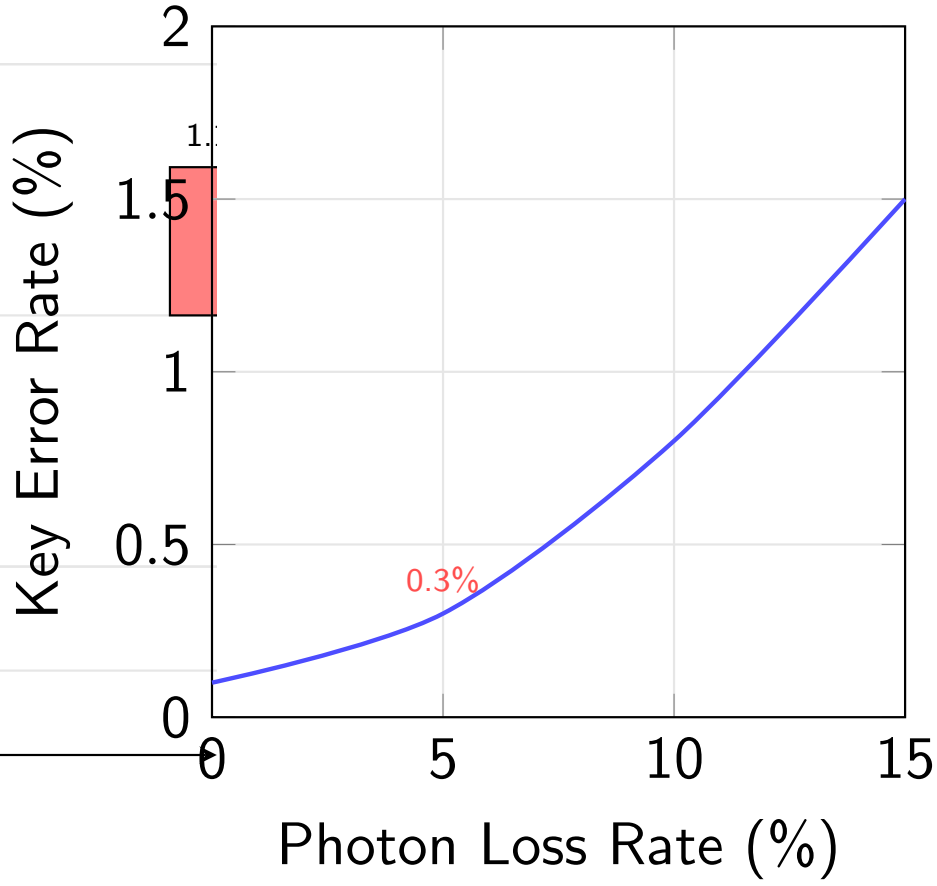

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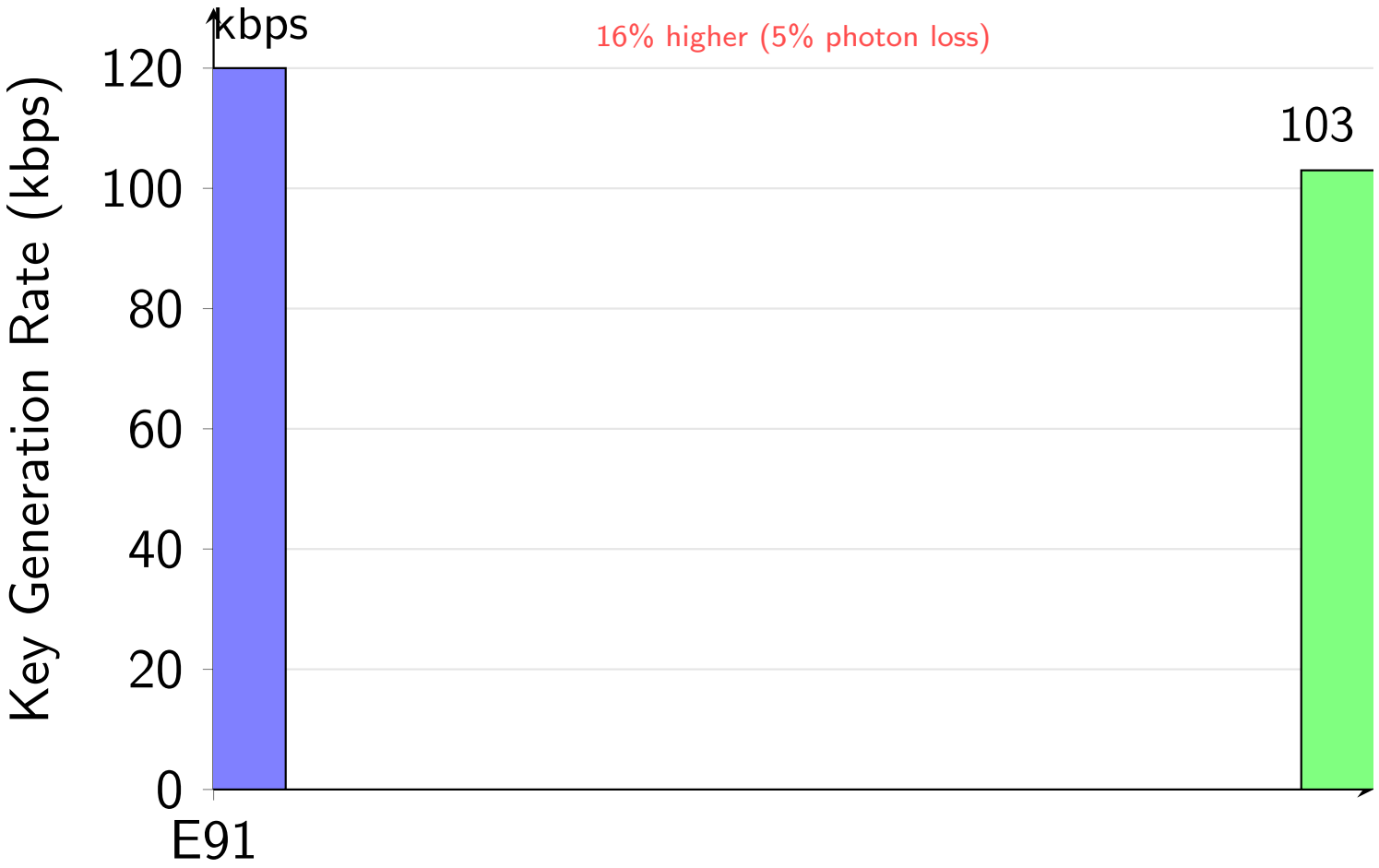

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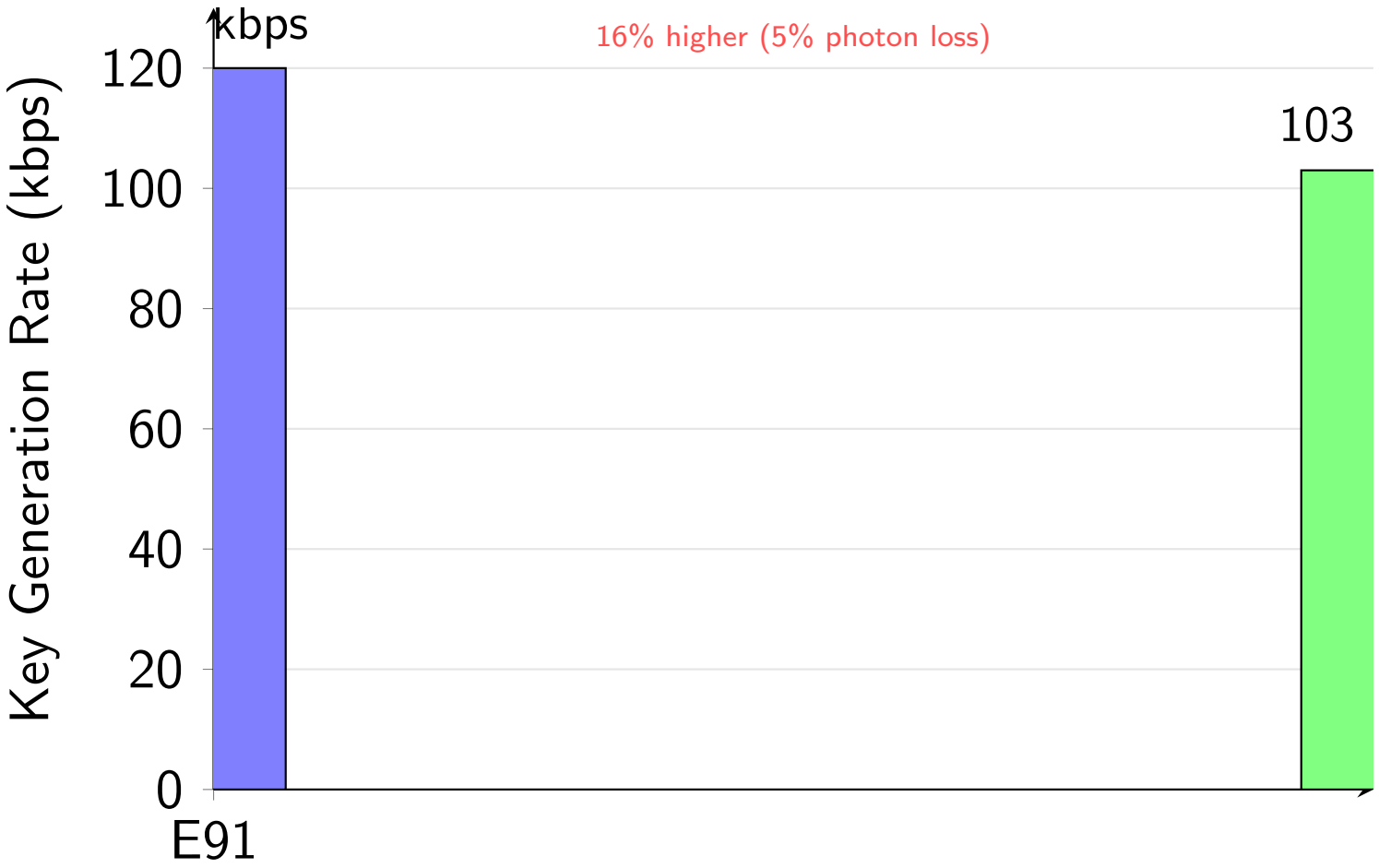

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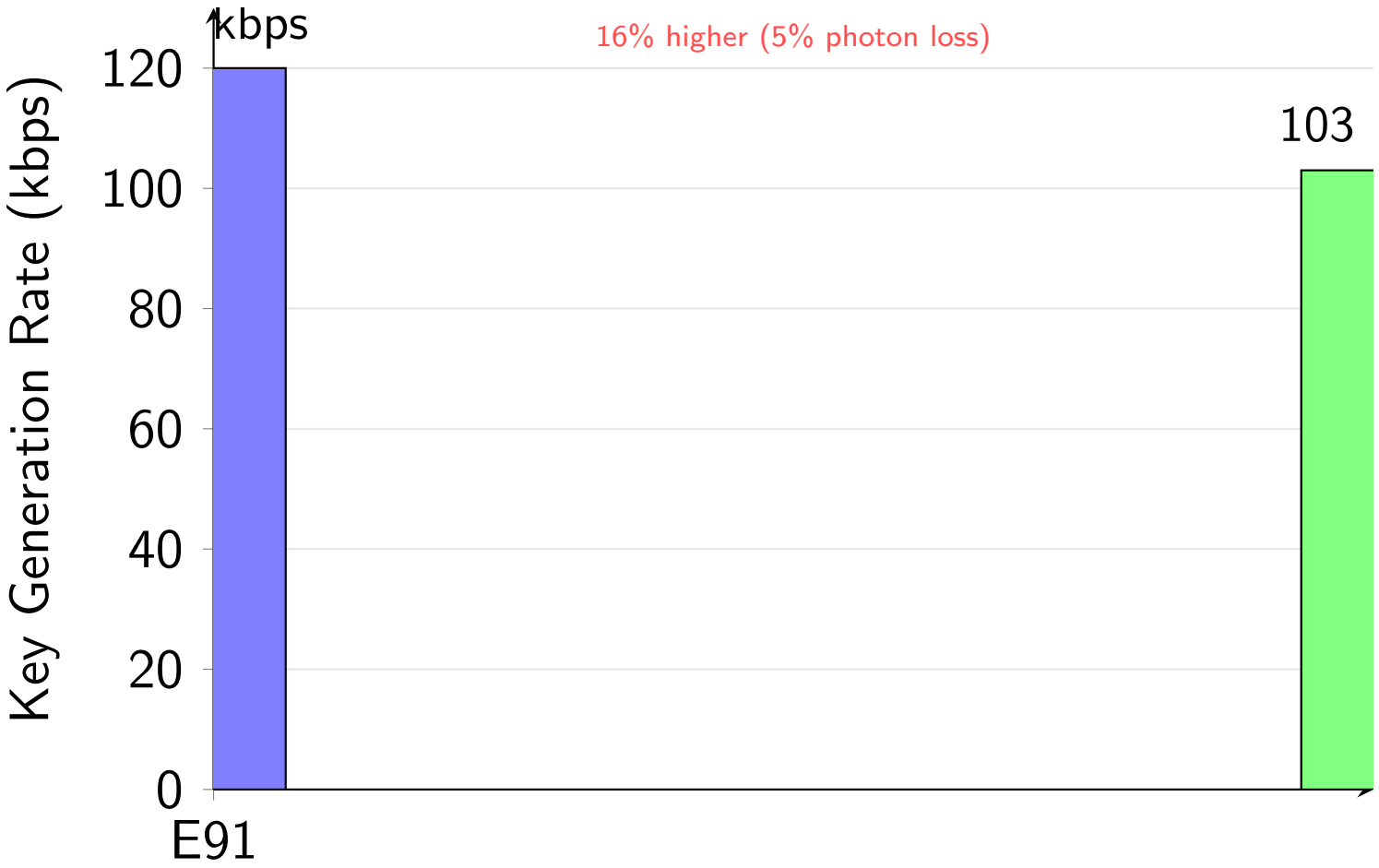

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- **Practical Trade-off:** BB84 is easier to implement; E91 excels in

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4. Future work 3: Integrate with MDI-QKD to further optimize security

Thank You for Listening! Any

Questions? Contact: [Your Email Address]